

PARKTOWN BOYS' HIGH SCHOOL



Department of Physical Science

Mid-Year Exam

June 2024

Subject	Physical Science	Marks	150
Grade	12	Time	3 hours
Topic	Physics		

GENERAL INSTRUCTIONS

1. Answer ALL the questions on your question paper.
2. The borrowing/lending of any material is FORBIDDEN.
3. Read the particular instructions for each section carefully (if applicable).
4. Answer all questions on the folio paper provided.
5. Marks may be forfeited if ALL instructions are not followed.
6. This test consists of 17 pages which includes this cover page.

Examiner

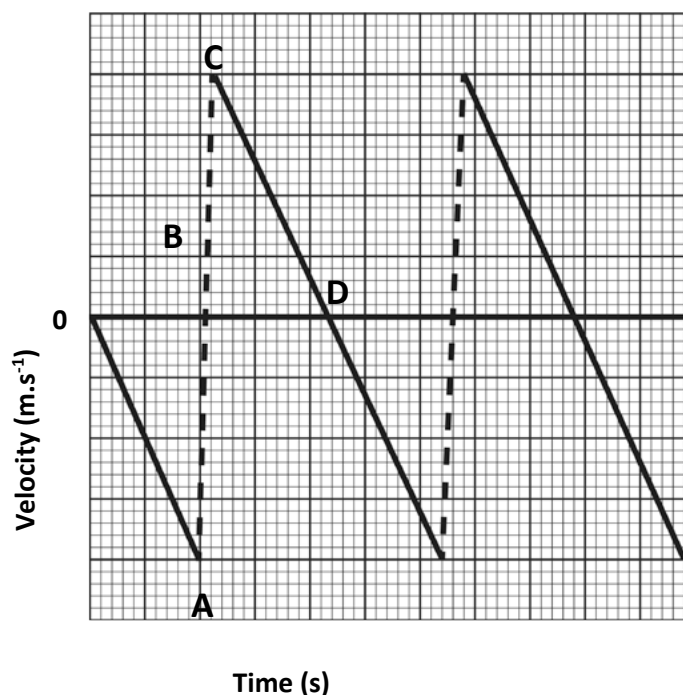
Ms L Whiting

Moderator

Mrs I Pelsner

Question 1**Multiple choice****[20]**

- 1.1 A ball is dropped from a certain height above the ground and bounces a few times as it hits the ground. The velocity-time graph below describes the motion of the ball from the time it is dropped. Ignore the effects of air friction.



In relation to the velocity-time graph for the bouncing ball shown above, which statement below is correct?

- A Down is taken as the positive direction and the ball is at the highest position of its first bounce at C.
- B Down is taken as the negative direction and the ball is at the highest position of its first bounce at D.
- C Down is taken as the negative direction and the ball is at the highest position of its first bounce at C.
- D Down is taken as the positive direction and the ball is at the highest position of its first bounce at D.

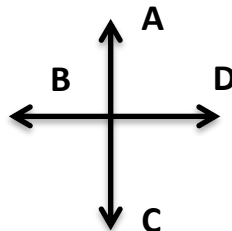
1.2 A diver of 50kg jumps from a platform 20m high into a pool. If the diver decelerates at a constant rate to zero velocity in 0.8 seconds after hitting the water, what is the force that the water exerts on the diver? Assume gravity in this question only as 10m.s^{-2}

- A 850N
- B 1680N
- C 1250N
- D 475N

1.3 In a certain electric field, the direction of the electric field is to the right as shown by the arrow below:



Which arrow below shows the direction an electron would move, if placed in this field?



1.4 A man is at rest in the middle of a smooth horizontal plane. He can move himself to shore making use of which Law?

- A Newtons 1st Law
- B Newtons 2nd Law
- C Newtons 3rd Law
- D Universal Law of Gravitation

- 1.5 Electric charges with magnitudes q , $2q$, $3q$ and $4q$ are placed in different electric fields. The force on each charge is measured and the results are recorded in the table shown below. Which charge experiences the greatest electric field strength?

	Magnitude of charge experiencing force	Force (N)
A	$4q$	55
B	$3q$	30
C	$2q$	25
D	q	20

- 1.6 Which of the following units is NOT a unit for power?

- A Watt
- B Kilowatt-hour
- C J.s^{-1}
- D N.m.s^{-1}

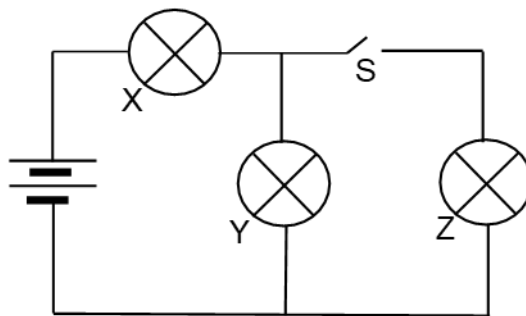
- 1.7 A heavy ball of mass $6m$ is travelling along a straight horizontal surface at a speed v . It collides with a stationary light ball of mass m . The surface on which they collide is frictionless. The heavy ball continues at half its original speed after the collision. The speed of the light ball after the collision will be equal to:

- A $2v$
- B $4v$
- C $3v$
- D $6v$

1.8 When a stone falls vertically, ignoring air friction,

- A there is no change in the total energy
- B E_k is conserved.
- C E_k decreases and gravitational E_p increases
- D E_p is conserved

1.9 The bulbs in this circuit are identical. The battery has negligible internal resistance.



When the switch is closed:

- A X and Y glow with the same brightness as before.
- B X and Y glow less brightly than before
- C X glows more brightly and Y glows less brightly.
- D X glows as brightly as before and Y glows less brightly

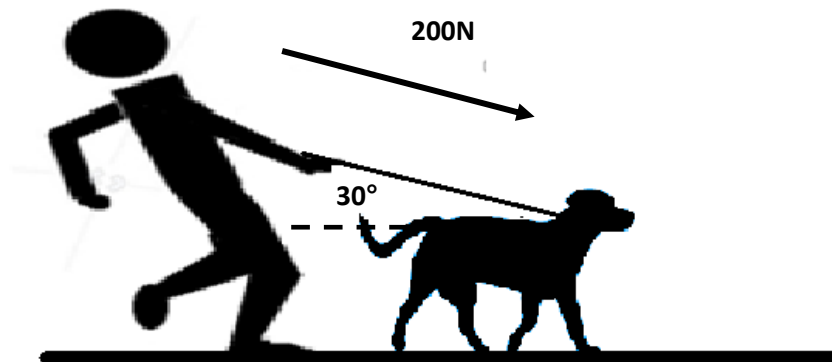
1.10 A kettle with a power of 2 000 W is used for a total of 20 minutes per day.
Assuming that the cost of electricity is R1,80 per kWh, the cost in rands, per day, of using the kettle is best expressed as:

- A $\frac{2 \times 20 \times 1,8}{1000}$
- B $\frac{2 \times 20 \times 180}{1000}$
- C $\frac{200 \times 20 \times 1,8}{1000 \times 60}$
- D $\frac{2 \times 20 \times 1,8}{60}$

Question 2

[10]

A child of mass 40 kg is holding a dog stationary when the dog pulls forward on an inextensible leash. The dog exerts a force of 200 N, so that the leash makes an angle of 30° to the horizontal as shown below.

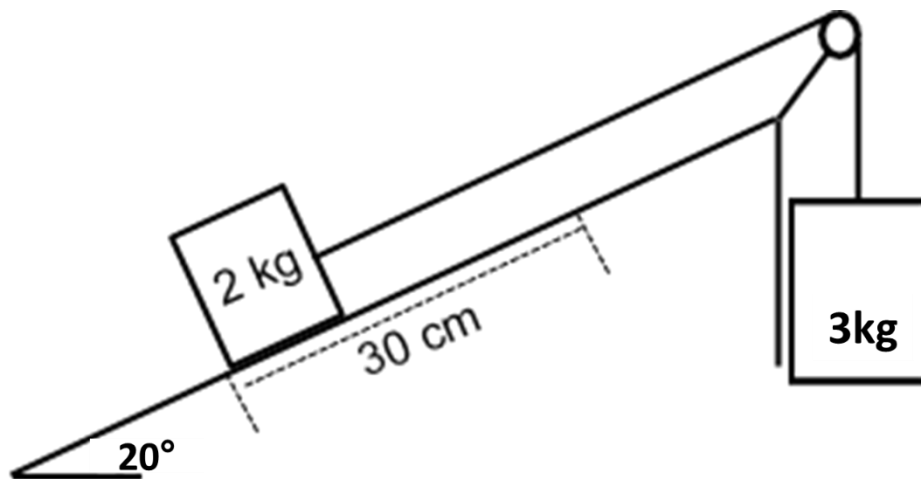


- 2.1 Calculate the magnitude of the normal force experienced by the child (4)
- 2.2 The coefficient of static friction between the child's foot and the ground is 0,4, calculate the maximum force of static friction experienced between the child's foot and the ground. (3)
- 2.3 Determine whether the child's foot will slip or not when the dog exerts the 200 N force at this angle and justify your answer. (3)

Question 3

[22]

A block of mass 3 kg is attached to a block of mass 2 kg via a light, inextensible string over a frictionless pulley as shown in the diagram. The 2 kg block is moving up the slope which is at an angle of 20° to the horizontal. The coefficient of kinetic friction between the block and the slope is 0,4.



- 3.1 Draw a labelled free-body diagram of the forces acting on the 2 kg block. (4)
- 3.2 Determine the magnitude of the normal force acting on the 2 kg block. (3)
- 3.3 Calculate the magnitude of the frictional force acting on the 2 kg block as it moves up the slope. (3)
- 3.4 Calculate the acceleration of the system AND the tension in the string. (6)

The 2 kg block moves 30 cm up the slope.

- 3.5 Calculate the energy lost due to the frictional force acting on the 2 kg block (3)
- 3.6 The string is **NOW** cut. What will the magnitude and direction of the acceleration of the 2 kg block be? (4)

Question 4

[21]

A stone with a mass of 300 g is dropped into a deep pool of water. Air resistance can be ignored. The stone hits the water with a velocity of $6 \text{ m}\cdot\text{s}^{-1}$ downwards. Once in the water, the stone slows down to a velocity of $0,8 \text{ m}\cdot\text{s}^{-1}$ downwards over a period of 0,2s. The stone then continues to sink at this velocity.

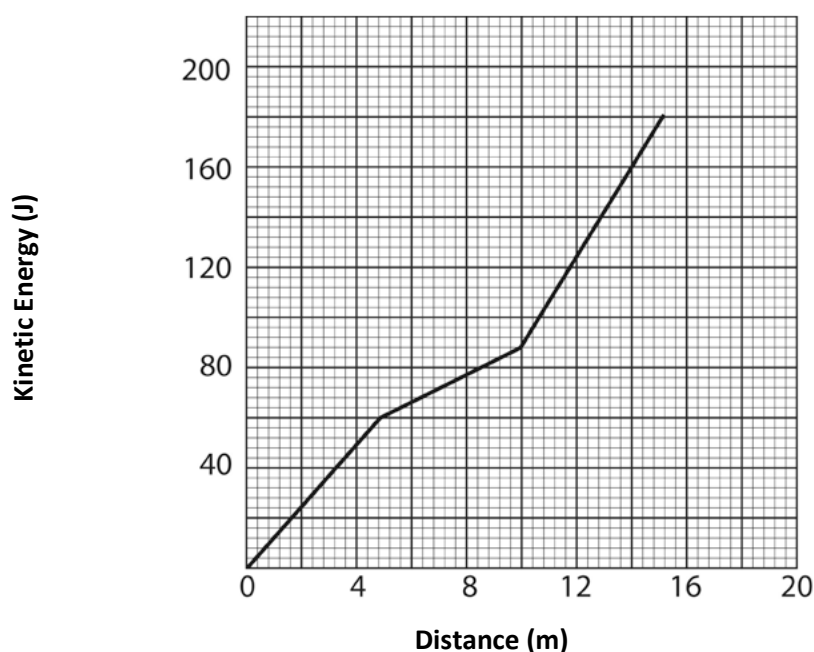
- 4.1 Calculate the height above the surface of the water from which the stone was dropped. (3)
- 4.2 How long did the stone take to fall, from when it was dropped to when it hit the water? (3)
- 4.3 Draw a velocity–time graph showing the velocity of the stone from the moment it was dropped to when it was sinking at constant velocity for 1,5 seconds. Indicate all the known time values. Assume a constant average acceleration for the stone after entering the water. (4)
- 4.4 Define impulse. (2)
- 4.5 Calculate the average net force experienced by the stone while it is slowing down in the water during the 0,2 seconds described. (4)
- 4.6 Determine the magnitude of the average upward force that the water exerts on the stone. (3)
- 4.7 Considering the forces acting on the stone, explain why it continues to fall at a constant $0,8 \text{ m}\cdot\text{s}^{-1}$ downwards shortly after landing in the water. (2)

Question 5

[20]

A box of **mass m** is pushed, from rest, with a constant applied force of 20 N over a horizontal surface which has three different regions, A, B and C. Each region is 5 m long and exerts a different frictional force on the box as it moves through. The regions are A, B and C as illustrated in the diagram below.

The graph below shows the kinetic energy of the object as a function of the distance from the starting position as the object experiences the constant applied force of 20 N.



- 5.1 State the work-energy theorem. (2)
- 5.2 Use the work-energy theorem to show that the gradient (slope) of the graph represents the net force experienced by the object. (3)
- 5.3 Calculate the net force experienced by the object in region C. (4)
- 5.4 Hence determine the magnitude of the friction force experienced by the block as it moves in region C. (3)

5.5 Which region, A, B or C offers the largest friction force? Justify your answer (3)

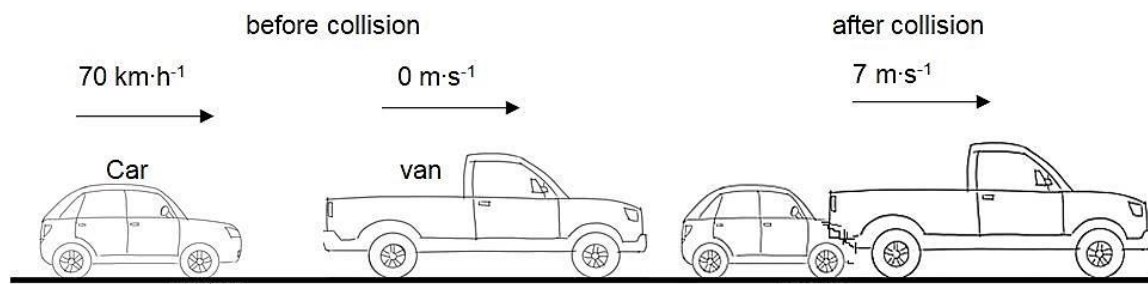
A car engine dissipates a power of 60 000 W while the car is travelling at a constant velocity of $60 \text{ km}\cdot\text{h}^{-1}$.

5.7 Define Power (2)

5.8 Calculate the force applied by the engine. (3)

Question 6 [8]

The car of mass 1 500 kg travelling to the right at $70 \text{ km}\cdot\text{h}^{-1}$ runs into the back of a stationary van of unknown mass. The two vehicles lock together and move to the right at $7 \text{ m}\cdot\text{s}^{-1}$.



6.1 State the law of conservation of momentum (2)

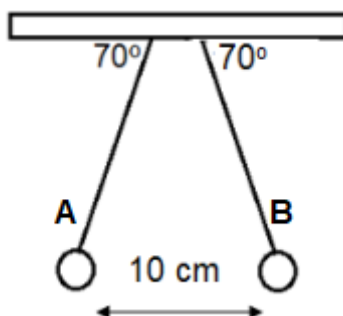
6.2 Calculate the mass of the van. (3)

6.3 Determine whether the kinetic energy is conserved or not (3)

Question 7

[22]

Two small identical polystyrene balls, A and B, each of mass 2 g are hung by insulating threads from a support. Each ball is given an identical negative charge. The balls repel each other so that they reach equilibrium 10 cm apart with the threads at angles of 70° to the horizontal as shown in the diagram below.



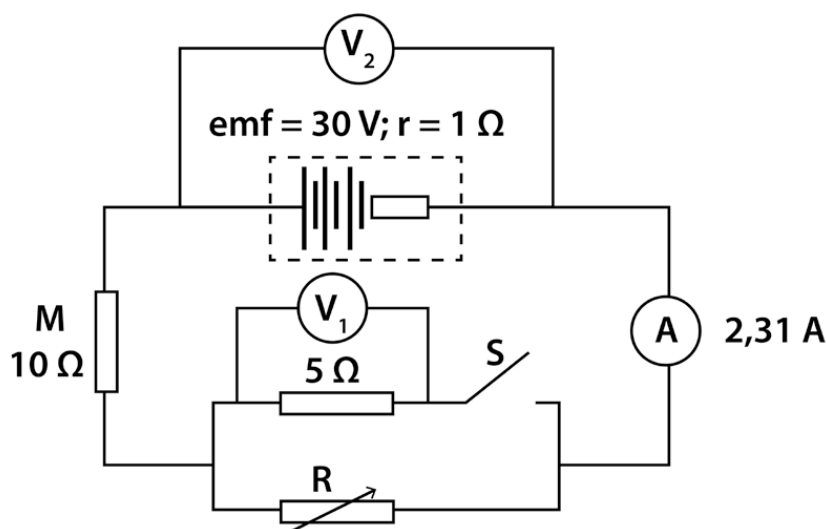
- 7.1 Indicate the electric field pattern around the two charges (2)
- 7.2 Calculate the weight of one of the polystyrene balls. (2)
- 7.3 Use a free body diagram and show all forces acting on Sphere A (3)
- 7.4 State Coulomb's law. (2)
- 7.5 Calculate the magnitude of the negative charge on each polystyrene ball. (6)
- 7.6 Calculate the electric field strength 30 cm away from Sphere B (3)
- 7.7 A man has a mass of 70kg. While on the planet Beeble, it is determined that it has a mass half of that of earth and a radius one quarter of the earth.
 - 7.7.1 State Newton's law of universal gravitation (2)
 - 7.7.2 Without using values, derive using formulas, the man's weight on Beeble. (5)

Question 8

[24]

A circuit consists of a battery with an emf of 30 V and an internal resistance of 1 Ω .

A motor M, a variable resistor (rheostat) which can be set at different values R, a 5 Ω resistor and a switch S are connected as shown in the diagram below



The reading on the ammeter is 2,31 A with the switch open:

- 8.1 Calculate the power dissipated in the motor M. (3)
- 8.2 Calculate the resistance at which the rheostat is set, R. (4)
- 8.3 What is the reading on the voltmeter V_1 ? (1)
- 8.4 What is the reading on the voltmeter V_2 ? (2)
- 8.5 Define resistance. (2)
- 8.6 Suppose the rheostat is now adjusted so that it has a higher resistance.
 - 8.6.1 Will the power dissipated in the motor increase, decrease or remain the same? (1)
 - 8.6.2 Justify your answer in Question 8.6.1 referring to two relevant formulae. (3)

8.7 The rheostat is adjusted so that its resistance is $10\ \Omega$ and the switch is now closed.

8.7.1 Calculate the reading on the ammeter. (5)

8.7.2 Calculate the reading on the voltmeter V1. (3)